TSP TENSILE STRUCTURES PAKISTAN

EDITION 2026/2027
OFFICIAL CORPORATE PROFILE

ARCHITECTURAL MEMBRANE DESIGN · STRUCTURAL STEEL FABRICATION · CANTILEVER SHADES



PREMIER ARCHITECTURAL MEMBRANE PRACTICE

Corporate Company Profile & Architectural Engineering Dossier

Turnkey design, non-linear form finding, computerized CNC patterning, German & French PVDF membrane fabrication, structural steel galvanizing, and site tensioning across Pakistan. Backed by 18+ years of engineering leadership.

18+ Years	120+ Completed	250,000+ Sq.Ft	15–25 Years	100% In-House
Engineering Pedigree	Landmark Projects	Covered Area Shaded	Fabric Life & Warranty	Design & Fabrication

HEAD OFFICE: DHA Phase 5, Lahore, Punjab
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ABOUT TSP TENSILE STRUCTURES

Executive Summary & Corporate Identity

TSP Tensile Structures is Pakistan’s foremost design-led tensile architecture and structural steel engineering firm. Headquartered in Lahore with regional desks in Islamabad and Karachi, TSP specializes in the complete turnkey delivery of architectural tensile fabric structures, cantilever car parking shade canopies, conical pavilions, curved walkways, and large clear-span PEB roofs.

Unlike conventional shade vendors who assemble prefabricated pipe frames, TSP operates as an integrated engineering practice. Every structure undergoes rigorous computational 3D form-finding, finite element structural analysis (FEA), wind pressure simulation under regional building codes (130 km/h wind gusts), automated CNC cutting patterning, and heavy-duty hot-air welded pre-stress tensioning.

OUR VISION	OUR MISSION
To elevate Pakistan's built environment by delivering iconic, long-lasting tensioned membrane structures that combine architectural beauty with 25-year structural permanence.	To eliminate sub-standard, low-grade shade systems in Pakistan by providing transparent European membrane materials, automated high-frequency welding, and certified turnkey engineering.

Four Pillars of Engineering Superiority

1. European Imported Membranes We exclusively partner with global leaders Mehler Valmex (Germany), Serge Ferrari (France), and Sioen (Belgium). Zero sub-standard Chinese tarpaulins.	**2. Grade-50 Steel & Hot-Dip Galvanizing** Heavy ASTM A500 circular tubular masts treated with 80–100 microns hot-dip zinc galvanizing. Maximum corrosion resistance in Pakistan’s climate.
3. In-House CNC & Hot-Air Welding Automated CAD membrane patterning with computerized hot-air and high-frequency welding machines. Seamless molecular lap seams that will never peel.	**4. Turnkey Single-Point Accountability** From initial 3D laser survey and concrete foundation casting to hydraulic tensioning and 15-year warranty certificate. No subcontractors.



Precision Rigging: SS-316 stainless steel turnbuckles and CNC clamps.



Commercial Canopy Delivery: Structural steel erection & tensioning.

EXECUTIVE LEADERSHIP

Founder & Chief Executive Profile

SHAUKAT RAUF

Founder & Chief Executive | Mechanical & Structural Engineering

18+ Years
Tier-1 EPC Experience

Shaukat Rauf is an esteemed Mechanical and Architectural Engineering professional with an illustrious 18+ year career spearheading complex megaprojects across Saudi Arabia and Pakistan. Prior to founding TSP Tensile Structures, he established a distinguished track record in tier-1 engineering and contracting environments including:

ORGANIZATION / EMPLOYER	LOCATION & SCOPE	ENGINEERING FOCUS
Haif Trading & Contracting	Riyadh, Kingdom of Saudi Arabia	Heavy structural steel fabrication, industrial tensioned canopies, defense & infrastructure megaprojects.
SAJCO (Shibh Al Jazira Contracting)	Saudi Arabia	Highways, highway toll plazas, architectural canopies, tensile toll station canopies.
Descon Engineering	Pakistan / Regional	Industrial steel structures, pressure vessels, precision rigging & equipment erection.
FWO (Frontier Works Organisation)	Pakistan	Strategic infrastructure, bridges, heavy steel trusses, fast-track turnkey execution.
Sarwar & Company	Pakistan	Commercial and institutional construction, specialized architectural facades & canopy engineering.

"In Saudi Arabia and international megaprojects, tensile membrane engineering is treated with the highest mathematical rigor. When I returned to Pakistan, I recognized a vast gap: clients were being sold weak, local sheet shades that sag within two years. At TSP, we brought international standards — imported European PVDF membranes, automated CNC patterning, high-tensile anchor engineering, and 15-year real warranties. When you invest with TSP, your structure will stand proud for a quarter century."

— Shaukat Rauf, Founder & Chief Executive, TSP Tensile Structures

CORE CAPABILITIES

Tensile Architectural Systems & Solutions

1. Cantilever Car Parking Shades	**2. Conical Umbrellas & Gazebos**
Single or double post cantilever designs with zero front-column obstruction. Allows effortless vehicle maneuvering for 1, 2, 3 or continuous commercial parking bays. 100% UV protection and 15°C temperature drop.	High-peak central mast tensile umbrella cones. Ideal for hotel courtyard dining, farmhouse lawns, rooftop cafes, and swimming pool lounges. Designed to handle intense downpours with integrated ring gutters.
3. Hyperbolic Paraboloid (Hypar) Sails	**4. Covered Pedestrian Walkways**
Sculptural 3D twisted surfaces with opposing anticlastic curvature. Creates dramatic visual statements for university campuses, luxury private residences, and corporate entrance atriums.	Linear continuous arched canopies with perimeter drainage for hospital corridors, international airports, golf clubs, and school pathways. Complete protection against monsoon rains and summer sun.
5. Rooftop Tensile Pergolas	**6. Industrial PEB & Sports Arenas**
High wind-load rated lightweight roof canopies anchored into concrete slabs via Hilti chemical anchors. Transforms unutilized terraces into premium corporate lounges and barbecue decks.	Large clear-span structural steel truss systems (30m to 60m span) covered with translucent architectural membranes. Perfect for padel courts, sports stadiums, logistics warehouses, and factory roofs.



Cantilever Car Shade: DHA Villa



Curved Walkway: Institutional Corridor



Luxury Pool Shade: Farmhouse Sail

MATERIAL SCIENCE

Imported Tensile Membranes & Technical Specs

A tensile structure's lifespan is entirely dictated by the chemical formulation of its architectural membrane. Low-grade local sheets deteriorate under UV exposure within 24 months, whereas imported European architectural membranes remain pristine for over 25 years. TSP exclusively utilizes tested, manufacturer-certified architectural membranes:

MEMBRANE SPECIFICATION	ORIGIN & GSM	FIRE & UV COATING	WARRANTY & LIFE
German Mehler Valmex PVDF ★ Flagship Gold Standard	Mehler Texnologies, Germany 900 GSM Architectural Grade	Self-cleaning dual-sided PVDF lacquer. DIN 4102 B1 flame retardant. 100% waterproof.	**15 Years Warranty** 20–25 Years Lifespan
French Serge Ferrari PVDF Précontraint Technology	Serge Ferrari, France 1050 GSM Heavy Industrial	Patented bi-directional pre-tensioning during coating. Zero elongation, extreme storm resistance.	**20 Years Warranty** 25–30 Years Lifespan
Architectural PTFE Glass Fibre Landmark Permanent Grade	Japan / Germany 1250 GSM Woven Glass	Class A1 non-combustible permanent fabric. 15% natural daylight transmission. Chemical inertness.	**25 Years Warranty** 35+ Years Lifespan
UV-Knitted HDPE Shade Cloth Breathable Micro-Perforated	South Korea / UAE 340 GSM High Density	95% UV blockage. Micro-perforated mesh allows hot trapped air to escape rapidly.	**8 Years Warranty** 10–12 Years Lifespan
Korean Commercial PVDF Cost-Effective Lacquer	South Korea 750 GSM Lacquered	100% waterproof acrylic & PVDF top-coat. Ideal for residential car porches and plazas.	**10 Years Warranty** 12–15 Years Lifespan

Hot-Air High-Frequency Lap Welding Technology

Tensile fabric panels cannot be stitched with standard needles, as sewing creates needle punctures that rip under high wind tension. At TSP, all membrane seams are fused using **computerized hot-air thermal fusion and high-frequency (HF) molecular welding**. The resulting lap seam is stronger than the parent fabric itself, guaranteeing 100% waterproof integrity and resistance to peeling under hurricane-force gusts.

STRUCTURAL INTEGRITY

Structural Steel Fabrication & Anti-Rust Systems

A tensile fabric membrane exerts massive pulling loads on its structural steel masts and perimeter catenary cables. To guarantee permanent structural stability, TSP fabricates framework using prime Grade-50 structural hollow sections and heavy-wall circular pipes.

COMPONENT	TSP ENGINEERING SPECIFICATION	BENEFIT & LONGEVITY
Primary Columns & Masts	ASTM A500 Grade-50 seamless / ERW circular hollow sections (8" to 14" dia, 6mm–12mm wall thickness).	Resists extreme bending moments under cantilever vehicle shade spans without deflection.
Anti-Corrosion Treatment	Hot-Dip Galvanizing (80–100 microns zinc bath conforming to ASTM A123 / ISO 1461).	Zero rust or corrosion even in humid coastal atmospheres (Karachi) and chemical rainfall.
Exterior Finish	Zinc chromate epoxy primer + 2 coats of oven-baked polyurethane powder coating or polyurethane enamel.	High-gloss or architectural matte aesthetic matching client villa / plaza color schemes.
Perimeter Wire Ropes	Marine Grade Stainless Steel (SS-316) 7x19 wire ropes (8mm–16mm dia) with swaged fork terminals.	Enclosed within fabric catenary pockets to distribute wind uplift evenly to corner anchor plates.
Tensioning Hardware	Drop-forged SS-316 turnbuckles, stainless steel D-shackles, and CNC machined clamp plates.	Allows micro-adjustment during hydraulic pre-stressing; zero seize or thread degradation.
Anchor Bolts & Footings	Grade 8.8 high-tensile galvanised anchor studs with chemical epoxy tie-ins or reinforced concrete pile caps.	Engineered against maximum overturning moments calculated by structural software.



Figure: Precision machined corner gusset plate with stainless steel rigging cable termination and hot-air welded catenary sleeve.

PROJECT EXECUTION

6-Stage Turnkey Engineering Workflow

TSP delivers seamless end-to-end execution with a single point of commercial responsibility. Our proven 6-stage engineering process ensures total accuracy from digital site survey to final pre-stress certification:

STAGE 1: 3D LASER SITE SURVEY & FEASIBILITY Our engineers visit your site with digital laser survey tools to measure boundary lines, driveway turning radii, underground utilities, and wind exposure orientations. Free nationwide survey booking.	**STAGE 2: 3D CAD FORM-FINDING & FEA SIMULATION** Using specialized tensile software (Forten / Easy / SAP2000), our design office performs non-linear shape form-finding. We optimize curvature for maximum rainwater drainage and wind pressure resistance (130 km/h).
STAGE 3: CNC PATTERNING & THERMAL WELDING 3D membrane surfaces are converted into 2D flattened cutting patterns with biaxial compensation factors. Fabric rolls are precision CNC-cut and fused with computerized hot-air welding equipment in our cleanroom facility.	**STAGE 4: HEAVY STEEL FABRICATION & GALVANIZING** Steel masts, anchor plates, and cantilever arms are fabricated in our Lahore workshop. Masts undergo full shot-blasting followed by 80-micron hot-dip galvanizing and oven-baked electrostatic powder coating.
STAGE 5: CIVIL FOUNDATIONS & MAST ERECTION Reinforced concrete footing cages are cast with Grade 8.8 anchor bolts, or chemical anchor tie-ins are installed onto existing reinforced concrete beams. Masts are lifted, plumb-aligned, and torque-fastened.	**STAGE 6: RIGGING, HYDRAULIC TENSIONING & HANDOVER** The prefabricated membrane is hoisted, perimeter SS-316 cables threaded, and hydraulic tensioners calibrated to design pre-stress levels. Handover includes 15-year warranty certificate and maintenance manual.



PROVEN TRACK RECORD

Landmark Projects: Commercial & Infrastructure

TSP has engineered signature tensile landmarks across Pakistan for multinational corporations, federal broadcast facilities, sports complexes, and private estates. Below is a curated selection of recent institutional deliveries:



RADIO PAKISTAN NATIONAL BROADCAST COMPLEX

Islamabad · Institutional Broadcast Facility

System: Continuous Arched Cantilever Canopy & Covered Walkways
Membrane: German Mehler Valmex PVDF (900 GSM)
Covered Area: 4,200 sq.ft covered parking & official entrance walkways
Features: Heavy hot-dip galvanized framework designed for severe wind exposure. Complete protection for broadcasting OB vans and executive vehicles.



INTERNATIONAL SPORTS ARENA & PADEL BLEACHERS

Lahore · Commercial Sports Complex

System: Large Clear-Span Tensile Membrane Bleacher Roof
Membrane: French Serge Ferrari Précontraint 1050 GSM
Covered Area: 6,500 sq.ft clear unobstructed spectator coverage
Features: High translucent fabric allows natural daylight during daytime matches while providing complete shade and acoustic buffering.

PROVEN TRACK RECORD

Landmark Projects: Residential & Luxury Estates



DHA PHASE 5 & PHASE 6 LUXURY VILLA CANOPIES

Lahore · Premium Residential Enclaves

System: Architectural Cantilever Car Porch Shade (2 & 3 Car Bays)
Membrane: German Mehler Valmex PVDF (900 GSM)
Covered Area: 450 to 900 sq.ft per residence
Features: Zero obstruction posts at driveway entrance. Oven-baked powder-coated steel posts with integrated LED lighting conduits.



EV CHARGING STATION & COMMERCIAL CORRIDOR

Islamabad Motorway Service Area

System: Double Cantilever Gull-Wing Canopy
Membrane: German Mehler Valmex PVDF (900 GSM)
Covered Area: 3,200 sq.ft fast charging bays
Features: Solar-ready engineering framing with rainwater collection troughs protecting high-voltage DC fast charging dispensers.

COMMERCIAL TRANSPARENCY

Turnkey Estimation & Pricing Matrix (PKR)

At TSP, we believe in radical commercial transparency. Our pricing includes structural steel fabrication, German/imported membrane, anchor bolts, foundation tie-ins, hydraulic tensioning, and warranty certificate with zero hidden mobilization extras.

FABRIC MEMBRANE OPTION	RATE (PKR / SQ. FT)	WARRANTY	BEST SUITED FOR
German Mehler Valmex PVDF (900 GSM)	**PKR 650 – 720**	15 Years	Premier residential car shades, commercial plazas, hotels.
French Serge Ferrari PVDF (1050 GSM)	**PKR 850 – 950**	20 Years	High wind coastal areas (Karachi), sports stadiums, heavy spans.
Architectural PTFE Glass Fibre	**PKR 1,550 – 1,800**	25 Years	Iconic airport terminals, major public transit landmarks.
Korean Commercial PVDF (750 GSM)	**PKR 460 – 520**	10 Years	Standard residential porches, school cafeterias.
Heavy UV Knitted HDPE Shade Cloth	**PKR 320 – 380**	8 Years	Cost-effective breathable parking lots, industrial yards.

Typical Turnkey Cantilever Car Parking Packages

1-Car Bay (10 x 18 ft = 180 sq.ft) Turnkey: PKR 115,000 – 130,000 German Mehler 900 GSM, Grade 50 steel post.	**2-Car Bay (20 x 18 ft = 360 sq.ft)** Turnkey: PKR 235,000 – 260,000 Most popular for executive homes & villas.
3-Car Bay (30 x 18 ft = 540 sq.ft) Turnkey: PKR 350,000 – 385,000 Double post cantilever with zero front column.	**10-Car Commercial (100 x 18 ft = 1800 sq.ft)** Turnkey: PKR 1,150,000 – 1,280,000 Corporate plazas, hospitals & banks.

Live Cost Estimation Tool: Clients can configure dimensions, fabrics, steel finishes, and cities instantaneously on our website (**www.tsptensile.com/calculator**) or via our official Mobile App.

INSTITUTIONAL TRUST

Quality Assurance, Codes & Client Endorsements

Strict Compliance with Global Structural & Fire Codes

ASTM A500 / A53: Structural steel tube & pipe standards.

ASTM A123 / ISO 1461: 80-micron hot-dip zinc galvanizing.

DIN 4102 B1 / NFPA 701: Flame retardant self-extinguishing fabric.

UBC 97 / ASCE 7-10: Structural wind load calculations up to 130 km/h.

Verified Client Testimonials

"We contracted TSP for our executive residential complex in DHA Lahore. The cantilever canopy execution was immaculate — no sag, zero water pooling even during intense monsoon downpours, and the German fabric quality is unmistakably superior to ordinary market shades. Engr. Shaukat Rauf's team is in a class of their own."

— Brig. (R) Tariq Mahmood, DHA Phase 5 Resident Committee

"As commercial architects, finding a tensile contractor in Pakistan with true non-linear engineering software capability was impossible until we found TSP. Their structural calculations, wind load compliance and clean cable terminations met international hospitality benchmarks."

— Ar. Kamran Siddiqui, Principal Architect, Studio Form & Space

"Our sports arena required a 6,500 sq.ft clear span membrane without internal pillars. TSP fabricated and erected the steel masts and Serge Ferrari membrane in just 22 days. Flawless workmanship and exemplary professionalism."

— Malik Usman Afzal, Managing Director, Arena Sports Complex

CONTACT & PROCUREMENT

Nationwide Office Directory & Direct Desk

TSP maintains fully mobilized engineering survey and rigging teams operating throughout Pakistan. Contact our central engineering coordination desk for complimentary laser site surveys, 3D conceptual concept renders, and detailed structural estimations:

CENTRAL HEADQUARTERS — LAHORE DHA Phase 5, Commercial Zone, Lahore, Punjab **Direct Phone:** 0302 4001063 **Office Tel:** 042-35789120 **Email:** info@tsptensile.pk **In-House Plant:** Sundar Industrial Estate, Lahore	**FEDERAL CAPITAL DESK — ISLAMABAD** F-8 Markaz, Executive Offices, Islamabad, ICT **Liaison Desk:** Saddar, Rawalpindi, Punjab **Direct Phone:** 0302 4001063 **Email:** islamabad@tsptensile.pk **Coverage:** Islamabad, Rawalpindi, Peshawar, Abbottabad
SOUTHERN REGIONAL DESK — KARACHI Clifton Block 4 / DHA Phase 6, Karachi, Sindh **Direct Phone:** 0302 4001063 **Email:** karachi@tsptensile.pk **Specialization:** Extreme coastal hot-dip galvanizing & marine grade SS-316 systems.	**DIGITAL PLATFORMS & MOBILE APP** **Official Website:** www.tsptensile.com **Cost Calculator:** www.tsptensile.com/calculator **Mobile App:** www.tsptensile.com/app **WhatsApp Instant Desk:** +92 302 4001063

READY TO COMMISSION YOUR TENSILE STRUCTURE?

Book your complimentary laser site survey today. Send photos or AutoCAD plots of your porch, courtyard, or commercial site via WhatsApp to **0302 4001063** to receive turnkey 3D concept renders within 24 hours.

TSP TENSILE STRUCTURES PAKISTAN — ENGINEERING ELEGANCE UNDER TENSION